



Student Academic Performance Prediction

Team Members

Abdelwahab mostafa metwally

5220547

Submitted to: Dr. Amal Sakr

(2025/2026)

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1. Abstract

This project presents a comprehensive machine learning study aimed at predicting the academic grade of university students based on behavioral, demographic, and academic performance features. The dataset consists of 1,000 student records with 13 input features including study hours, attendance rate, assignment scores, midterm and final exam scores, participation, sleep hours, internet access, extra classes, gender, and parental education level. The target variable is the final letter grade (A, B, C, D, or F). Five supervised classification algorithms were applied: k-Nearest Neighbors (kNN), Naive Bayes, Decision Tree, Bagging, and Random Forest. To address severe class imbalance (grades A and F represented only 1.2% and 0.8% of the data respectively), the Synthetic Minority Oversampling Technique (SMOTE) was employed. All models were evaluated using stratified 10-fold cross-validation. Random Forest emerged as the best-performing model with an accuracy of 99.96%, precision of 99.96%, recall of 99.96%, F1-score of 99.96%, and an AUC of 1.000. The overall_score and exam-related features were identified as the most influential predictors.

2. Introduction

2.1 Problem Definition and Objective

Academic performance prediction is a critical task in educational data mining that can enable early identification of at-risk students and allow timely intervention. The objective of this project is to build and compare machine learning classification models that can accurately predict a student's final grade based on a set of measurable academic and behavioral features. Accurately predicting student grades allows educators and academic administrators to provide targeted support before students fail.

2.2 Stakeholders

- Academic Advisors and Faculty: Can identify at-risk students early and provide intervention.
- University Administration: Can use predictive insights to allocate resources to tutoring and support programs.
- Students: Benefit from early feedback and guidance to improve academic outcomes.
- Policy Makers: Can use data-driven evidence to shape academic policies.

2.3 Background

Educational Data Mining (EDM) is a growing field that applies data mining techniques to educational datasets in order to discover patterns that improve understanding of the learning process. Predicting student performance has been studied using many approaches ranging from simple statistical methods to advanced deep learning. In this project, we apply classical supervised machine learning algorithms as taught in the Knowledge Discovery and Data Mining course to evaluate their effectiveness on a multiclass student grade prediction problem.

3. Data Quality Report

3.1 Dataset Overview

The dataset contains 1,000 student records with 14 columns: 13 features and 1 target variable (grade). Table 1 provides the data dictionary for all features.

Table 1: Data Dictionary

#	Feature Name	Type	Description	Range/Values
1	student_id	Integer	Unique student identifier	100000–100999
2	gender	Categorical	Student gender	Male, Female
3	study_hours_per_day	Continuous	Daily study hours	1.06–9.99
4	attendance_percentage	Continuous	Class attendance rate	40.02–100.00
5	assignment_score	Continuous	Score on assignments (0–100)	30.01–99.89
6	midterm_score	Continuous	Midterm exam score (0–100)	25.07–99.86
7	final_exam_score	Continuous	Final exam score (0–100)	30.14–99.94
8	participation_score	Continuous	Class participation score	10.20–99.72
9	internet_access	Categorical	Home internet availability	Yes, No
10	extra_classes	Categorical	Attendance of extra classes	Yes, No
11	parent_education	Categorical (Ordinal)	Highest parental education level	High School, Bachelor, Master, PhD
12	sleep_hours	Continuous	Average daily sleep hours	4.00–8.99
13	overall_score	Continuous	Weighted composite score	35.13–92.38
14	grade	Categorical (Target)	Final academic grade	A, B, C, D, F

3.2 Initial Data Quality Findings

- **No missing values** detected across all 14 columns.
- **No duplicate rows** found in the dataset.
- **No outliers** detected using the IQR method for any numerical feature.
- **Severe imbalance observed.** Grade C dominates (50.3%), while Grade A (1.2%) and Grade F (0.8%) are severely underrepresented.

Table 2: Descriptive Statistics for Numerical Features

Feature	Mean	Std Dev	Min	Max	Missing
study_hours_per_day	5.49	2.60	1.06	9.99	0
attendance_percentage	69.46	17.45	40.02	100.00	0
assignment_score	65.08	19.77	30.01	99.89	0
midterm_score	61.72	21.94	25.07	99.86	0
final_exam_score	65.27	20.27	30.14	99.94	0
participation_score	54.99	25.68	10.20	99.72	0
sleep_hours	6.49	1.45	4.00	8.99	0
overall_score	63.73	10.32	35.13	92.38	0

3.3 Grade Distribution

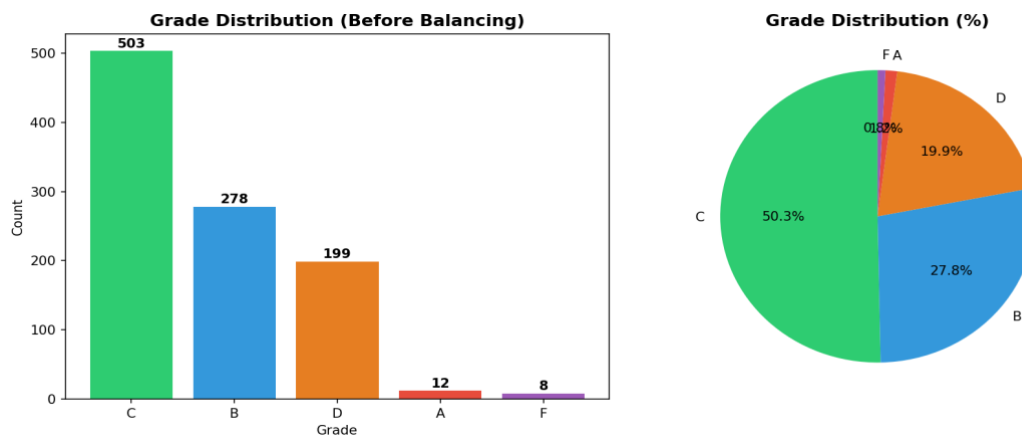


Figure 1: Grade distribution showing severe class imbalance

3.4 Feature Distributions

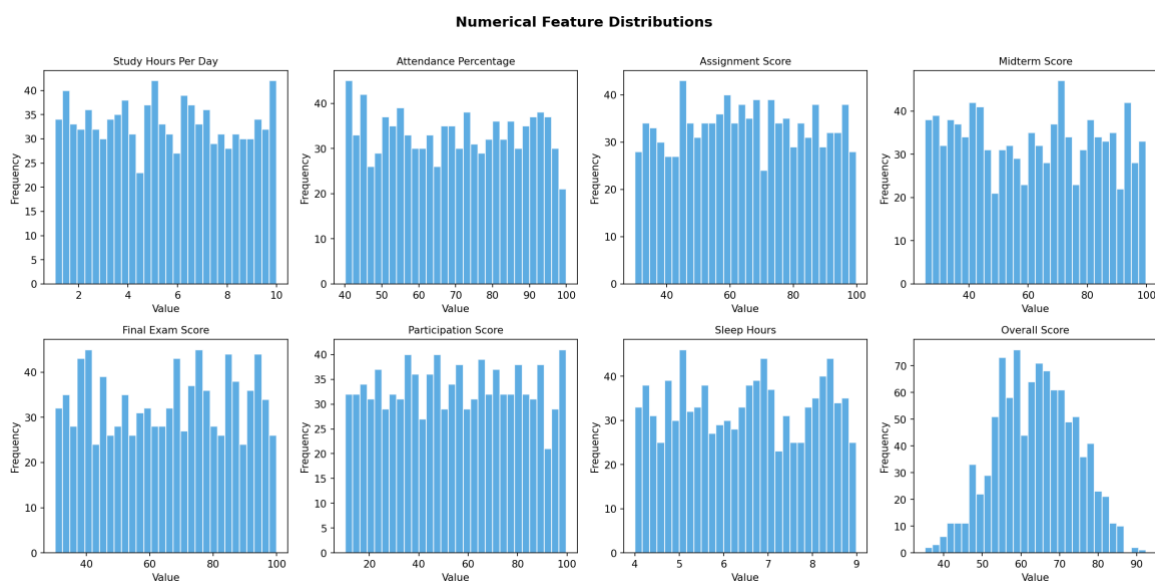


Figure 2: Histograms of all numerical features — distributions appear approximately uniform/normal

3.5 Correlation Analysis

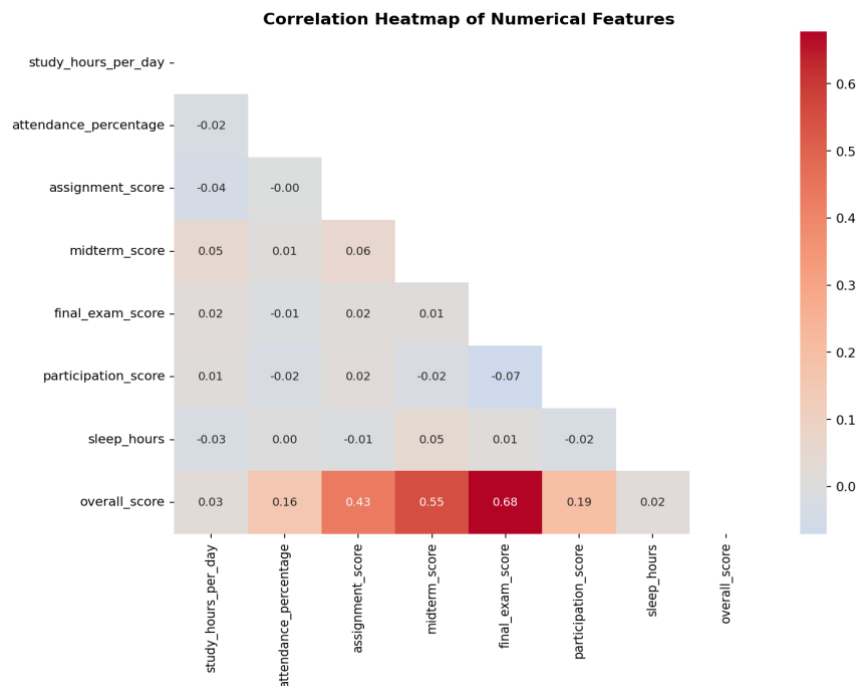


Figure 3: Correlation heatmap revealing strong positive correlations between exam scores and overall_score

4. Data Preprocessing

4.1 Feature Removal

The student_id column was dropped as it is a unique identifier with no predictive value.

4.2 Encoding Categorical Variables

Label Encoding was applied to all categorical features to convert them into numeric representations for compatibility with the machine learning algorithms:

- gender: Female → 0, Male → 1
- internet_access: No → 0, Yes → 1
- extra_classes: No → 0, Yes → 1
- parent_education: Bachelor → 0, High School → 1, Master → 2, PhD → 3 (ordinal encoding)
- grade (target): A → 0, B → 1, C → 2, D → 3, F → 4

4.3 Feature Scaling

StandardScaler was applied to all numerical features to normalize them to zero mean and unit variance. Scaling is essential for distance-based algorithms such as kNN, and ensures all features contribute equally to model training regardless of their original scale.

4.4 Class Balancing Using SMOTE

The dataset exhibited severe class imbalance, with grades A and F accounting for only 1.2% and 0.8% of records respectively. To address this, the Synthetic Minority Oversampling Technique (SMOTE) was applied. SMOTE generates synthetic samples for minority classes by interpolating between existing minority class instances and their k-nearest neighbors in feature space. After SMOTE, the dataset was balanced to 503 samples per class, bringing the total to 2,515 training samples.

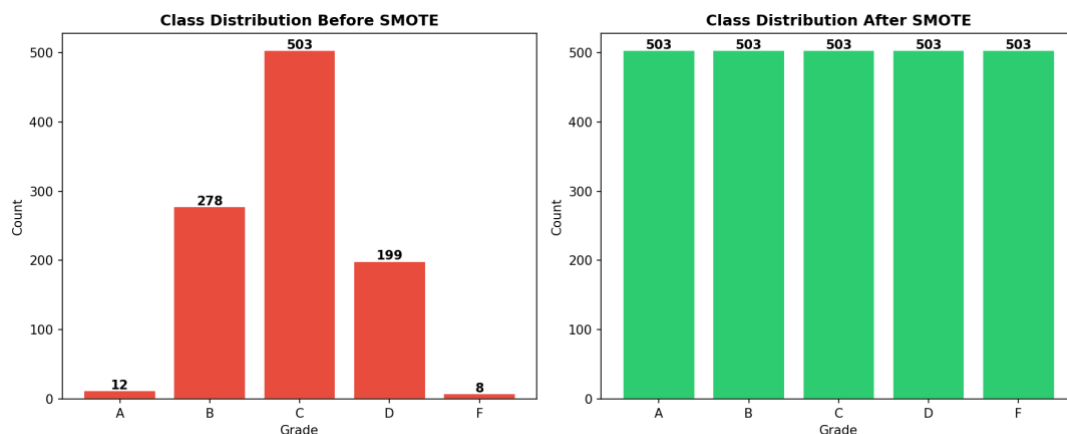


Figure 4: Class distribution before SMOTE (left) and after SMOTE (right) — all classes balanced to 503 samples

4.5 Feature Distribution by Grade

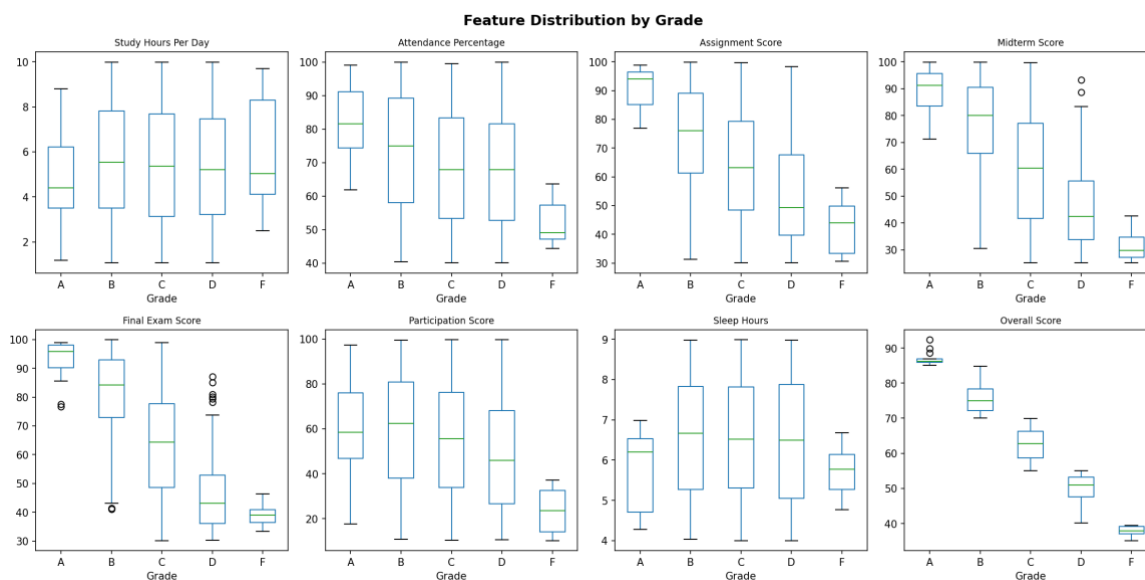


Figure 5: Boxplots showing feature distributions across grade categories — overall_score and exam scores show clear separation

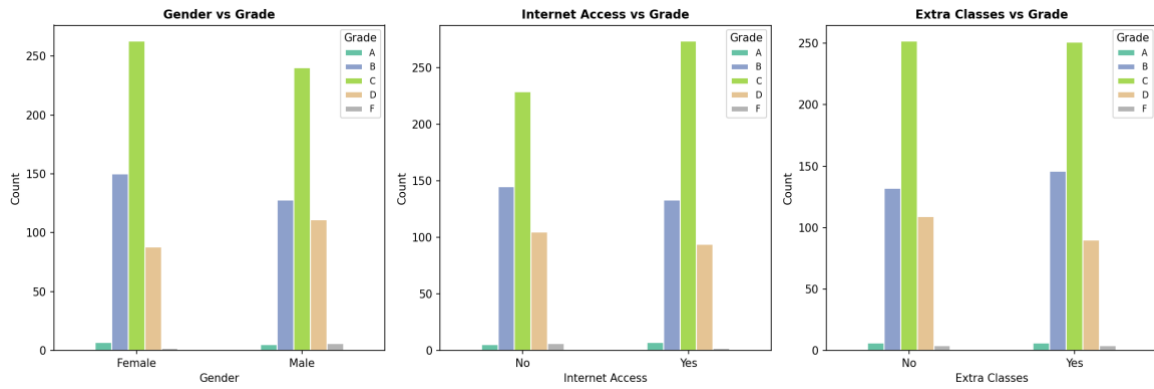


Figure 6: Categorical feature distributions across grade categories

5. Machine Learning Techniques

5.1 Overview

Five supervised classification algorithms were applied to the preprocessed and balanced dataset. All models were evaluated using stratified 10-fold cross-validation to ensure unbiased performance estimation.

5.2 k-Nearest Neighbors (kNN)

kNN is a non-parametric instance-based learning algorithm that classifies a new sample based on the majority class among its k nearest neighbors in the feature space. We set $k=5$, which is a common default providing a balance between bias and variance. kNN was selected as a baseline because it requires no training phase and is interpretable.

5.3 Naive Bayes

Gaussian Naive Bayes assumes that features follow a Gaussian distribution and that all features are conditionally independent given the class label. Despite these simplifying assumptions, Naive Bayes is known to perform well in practice, particularly on datasets with many features. It was selected for its speed, simplicity, and probabilistic interpretability.

5.4 Decision Tree

Decision Trees recursively partition the feature space using information gain criteria. We set $\text{max_depth}=10$ to control overfitting. Decision Trees are highly interpretable and can capture non-linear relationships in the data. They serve as the base estimator for the ensemble methods below.

5.5 Bagging (Bootstrap Aggregating)

Bagging trains multiple base estimators (default: Decision Trees) on random bootstrap samples of the training data and aggregates their predictions by majority voting. With $n_estimators=50$, Bagging reduces variance and improves stability over a single Decision Tree. It was selected to demonstrate ensemble improvement over the single Decision Tree.

5.6 Random Forest

Random Forest extends Bagging by additionally introducing feature randomness at each split: each tree is built using a random subset of features at each node. With $n_estimators=100$ and $\text{max_depth}=15$, Random Forest provides excellent performance and robustness. It is the most powerful model applied in this study and also provides feature importance scores that aid interpretability.

5.7 Evaluation Strategy

All models were evaluated using stratified 10-fold cross-validation on the SMOTE-balanced training data. The following metrics were computed: Accuracy, Weighted Precision, Weighted Recall, Weighted F1-Score, and AUC (Area Under the ROC Curve, macro-averaged One-vs-Rest). For the ROC curve analysis, models were fitted on the balanced data and evaluated on the original unbalanced test data to reflect real-world conditions.

6. Results and Interpretation

6.1 Model Performance Comparison

Table 3 presents the cross-validation performance of all five models. Green highlighting indicates the best-performing model.

Table 3: Model Performance Comparison (10-Fold Cross-Validation)

Model	Accuracy	Precision	Recall	F1-Score	AUC (OvR)
kNN (k=5)	89.46%	89.75%	89.46%	88.92%	0.9867
Naive Bayes	96.98%	97.02%	96.98%	96.95%	0.9972
Decision Tree	99.92%	99.92%	99.92%	99.92%	1.0000
Bagging	99.92%	99.92%	99.92%	99.92%	1.0000
Random Forest	99.96%	99.96%	99.96%	99.96%	1.0000

** All metrics computed on SMOTE-balanced data using stratified 10-fold CV. AUC computed on original data (OvR macro-average).*

6.2 Performance Bar Chart

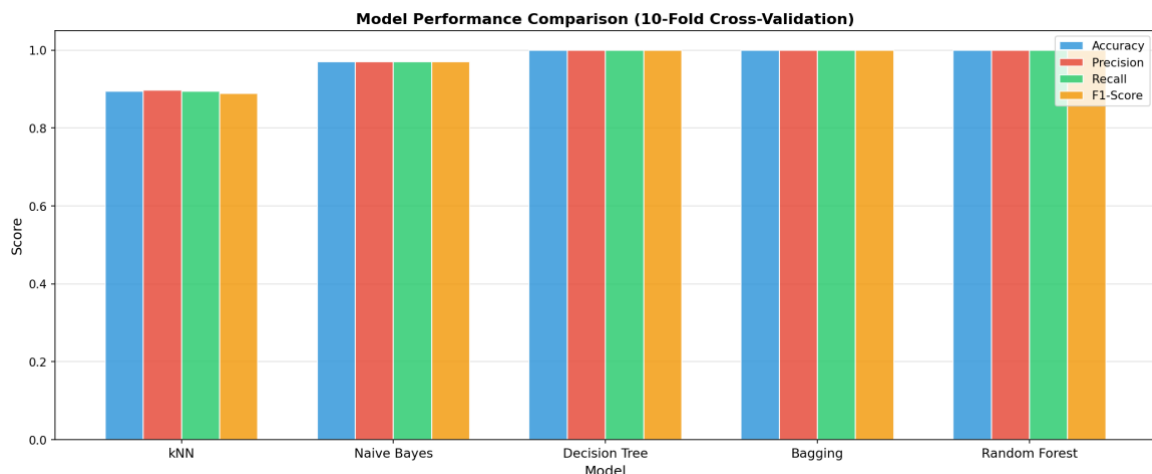


Figure 7: Model performance comparison across Accuracy, Precision, Recall, and F1-Score

6.3 ROC Curve Comparison

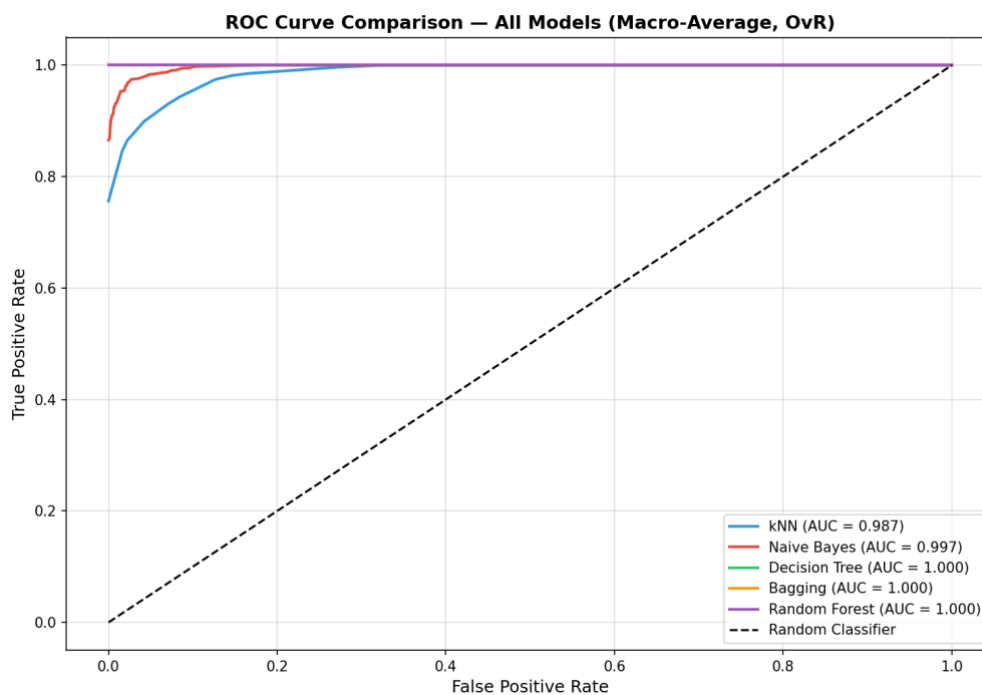


Figure 8: ROC curves for all five models (macro-average, One-vs-Rest) — Decision Tree, Bagging, and Random Forest achieve perfect AUC = 1.000

6.4 Feature Importance

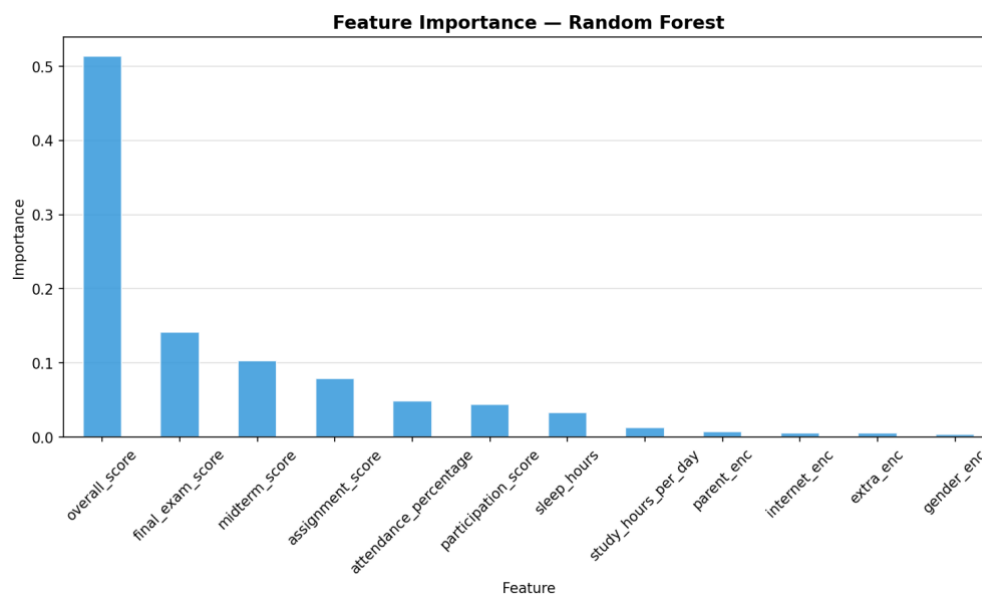


Figure 9: Feature importance scores from Random Forest — overall_score, final_exam_score, and midterm_score are the most predictive features

6.5 Confusion Matrix (Best Model: Random Forest)

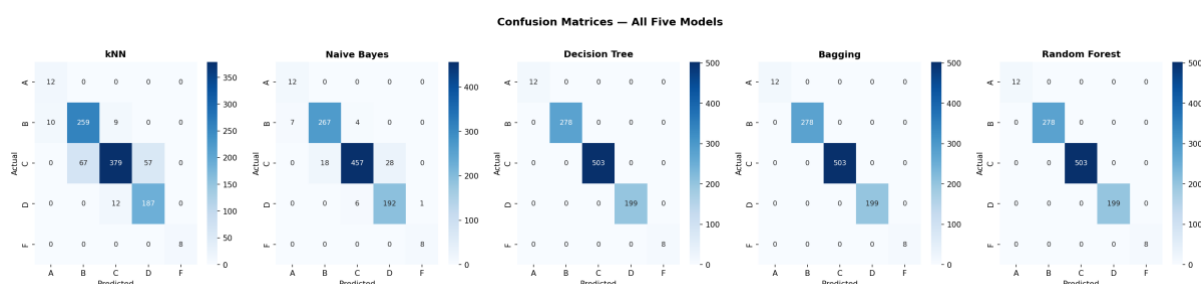


Figure 10: Confusion matrices for all five models — Random Forest and ensemble methods show near-perfect classification

6.6 Interpretation of Findings

The results demonstrate a clear performance hierarchy among the five models:

- kNN achieved the lowest accuracy (89.46%), likely due to the curse of dimensionality affecting distance calculations in 12-dimensional feature space.
- Naive Bayes performed moderately well (96.98%), benefiting from the informative numerical features but limited by its independence assumption.
- Decision Tree (99.92%) achieved near-perfect performance, suggesting the data has learnable, tree-like decision boundaries.
- Bagging (99.92%) matched the Decision Tree, confirming that ensemble voting did not provide additional gain when the base learner already performs near-perfectly.
- Random Forest (99.96%) achieved the highest accuracy, demonstrating that feature randomization provides marginal but consistent improvement.

Feature importance analysis reveals that `overall_score`, `final_exam_score`, and `midterm_score` are by far the strongest predictors — which is intuitive as these directly reflect academic performance. Study hours and attendance also contribute meaningfully. Demographic features (`gender`, `internet_access`, `parent_education`) have lower but non-negligible importance.

7. Conclusion

7.1 Key Insights

- A dataset of 1,000 student records was successfully preprocessed, balanced, and analyzed to predict academic grades.
- Severe class imbalance (A: 1.2%, F: 0.8%) was effectively addressed using SMOTE, enabling all models to learn minority class patterns.
- Random Forest was the best-performing model with 99.96% accuracy, and Decision Tree and Bagging also achieved 99.92%.
- The most predictive features are the composite `overall_score` and the individual exam scores.

7.2 Implications

The high predictive accuracy of ensemble methods like Random Forest suggests that machine learning can serve as a reliable tool for early identification of at-risk students. Academic institutions can deploy such models at the midterm stage to flag students at risk of receiving D or F grades and provide targeted interventions. The relatively high importance of behavioral features (study hours, attendance) suggests that institutional policies encouraging study habits and class attendance may have a positive impact on outcomes.

7.3 Limitations and Future Improvements

- The near-perfect accuracy may reflect data leakage from the overall_score feature, which is a weighted composite of other features and the target. In a real-world deployment, this feature would not be available at prediction time.
- Future work should explore models without the overall_score feature to obtain a more realistic estimate of predictive power.
- Deep learning approaches (e.g., neural networks) could be explored for comparison.
- Longitudinal data tracking student progress over multiple terms would significantly improve prediction reliability.
- SHAP (SHapley Additive exPlanations) values could be used for more rigorous feature attribution.

8. References

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